

Julian C. Stamm

M.Sc. Computer Science · Real-Time GI & Neural Rendering

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EDUCATION

- Oct 2025 – present **University of Bonn**, M.Sc. in Computer Science
 - Focus on Real-Time GI, Neural Rendering, and Machine Learning
- Oct 2021 – Sept 2025 **University of Bonn**, B.Sc. in Computer Science
 - Grade: 1.1 ([top 3%](#) )
 - Bachelor's Thesis (1.0): Bidirectional Neural Radiance Caching [[full-res](#) ] [[low-res](#) ] [[slides](#) )

PUBLICATIONS

- May 2026 **Photon-Driven Neural Radiance Caching** 

Introduces a new photon-based training method for Neural Radiance Caching, reducing FLIP error by up to two-thirds in caustic scenes with only a ~2 ms frame-time overhead. A continuation of my Bachelor's thesis, presented at I3D 2026.
 Julian C. Stamm, Tom Kneiphof, Reinhard Klein
[10.1145/3807895.3807923](https://doi.org/10.1145/3807895.3807923)  (I3D Companion 2026)

EXPERIENCE

- Feb 2026 – Mar 2026 **Teaching Assistant for "Foundations of 4D/6D Object Capture for Virtual Environments"**, University of Bonn – Bonn, Germany
 - Held daily block tutorials on NeRFs, IK, 3DGS, differentiable rendering, etc.
- Apr 2024 – present **Teaching Assistant for "Introduction to Computer Graphics"**, University of Bonn – Bonn, Germany
 - Held weekly tutorials on rasterization, ray tracing, BSDFs, splines, etc.
 - Assisted in creating exercises and exams
 - Developed and maintained a modern OpenGL/C++ framework used in the exercises and the animation competition
- May 2021 – Sept 2021 **Temporary Worker in Software Development Team**, Atlas Spare Parts GmbH – Ganderkesee, Germany
 Developed custom CAN bus logger.

PROJECTS

- May 2026 – Sept 2026 **Lab: Towards Real-Time Neural Path Guiding**
 Explored current state and real-time applicability of neural path guiding using Slang
[\[report\]](#)  [\[slides\]](#) 

May 2026 – present [tsnn](#) ↗

High-performance Slang-native neural density estimation using Slang's auto-differentiation.

- Implemented neural spline flows, neural mixture models, hash grid encodings, MLPs, Adam, etc
- Achieves full kernel fusion with Falcor path tracer
- Uses cooperative vectorization to accelerate MLP evaluation and backpropagation
- Outperforms tiny-cuda-nn with JIT in image-compression benchmark
- Written for my lab on real-time neural path guiding

Feb 2025 – May 2026 [PhotonNRC](#) ↗

Falcor implementation of the poster "Photon-Driven Neural Radiance Caching".

- Rebuilt the method as a clean Falcor render graph, improving modularity and performance
- Implemented SPPM with hardware-accelerated inverse radius search and stochastic accumulation, NRC via tiny-cuda-nn, and SPPC training

Feb 2025 – Sept 2025 [BiNRC](#) ↗

OptiX implementation of my Bachelor's thesis "Bidirectional Neural Radiance Caching".

- Built an unbiased path tracer with MIS and NEE and a basic Disney principled BSDF
- Implemented SPPM with hardware-accelerated inverse radius search
- Implemented NRC via tiny-cuda-nn with bidirectional and photon-driven (Photon-NRC) training

Aug 2024 – present [nBounce](#) ↗

Personal real-time path tracer with software BVH construction and traversal.

- Written in Rust and wgpu
- SAH-based BVH construction, Owen-scrambled RQMC sampling, VNDF importance sampling

Oct 2023 – Mar 2024 [Stable Real-Time Simulation of Mesoscopic Glints](#) ↗

Computer Graphics Lab, part of my B.Sc. in Computer Science.

- Implemented the discrete stochastic microfacet model from "Real-Time Rendering of Glinty Appearances Using Distributed Binomial Laws on Anisotropic Grids" by Thomas Deliot and Laurent Belcour

Aug 2023 – present [gltemplate](#) ↗

OpenGL 4.6 framework developed for CG exercises.

- Written in C++ and CMake, RAII-based resource management, texture/mesh loading utilities

OTHERS

Apr 2026 – present [bevy_trail](#) ↗

Work-in-progress bevy plugin for high-performance GPU-based trails in wgpu.

Sept 2025 – present **P2P Multiplayer Game**

Working on a Python-moddable P2P-multiplayer card game in Rust using bevy.

June 2023 – Sept 2023 [monarch-tree](#) ↗

Procedural tree and butterfly-swarm animation for the course "Introduction to Computer Graphics"

- Growing the tree with a self-devised model that steers branches using boids-like forces (up, spread, forward, noise).

SKILLS

Languages: C++, CMake, GLSL, CUDA, Slang, Python, Rust, WGSL, Java

APIs: OpenGL, OptiX, Falcor, wgpu, bevy, Godot, PyTorch

Research Interests: Real-Time GI, Neural Path Guiding, Neural Radiance Caching, Photon Mapping, Neural Density Estimation

Spoken Languages: German (native), English (TOEFL iBT: 110/120)